MORPHISMS OF ALGEBRAIC SPACES INDUCE MORPHISMS OF LOCALLY RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 04HA: sources/stacks-chow/tags.txt, source lines 5861–5861. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04KH: sources/stacks-chow/tags.txt, source lines 5973–5973. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original theorem 04K5: sources/stacks-chow/tags.txt, source lines 5961–5961. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04H9: sources/stacks-chow/tags.txt, source lines 5860–5860. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04K6: sources/stacks-chow/tags.txt, source lines 5962–5962. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02X1: sources/stacks-chow/tags.txt, source lines 3824–3824. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04JX: sources/stacks-chow/tags.txt, source lines 5953–5953. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 04I5: sources/stacks-chow/tags.txt, source lines 5891–5891. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04KF: sources/stacks-chow/tags.txt, source lines 5971–5971. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The morphism of ringed topoi $(f_{small}, f^\sharp)$ associated to f is a morphism of locally ringed topoi, see Modules on Sites, Definition 04HA.*

Proof. Note that the assertion makes sense since we have seen that $(X_{étale}, \mathcal{O}_{X_{étale}})$ and $(Y_{étale}, \mathcal{O}_{Y_{étale}})$ are locally ringed sites, see Lemma 04KH. Moreover, we know that $X_{étale}$ has enough points, see Theorem 04K5. Hence it suffices to prove that $(f_{small}, f^\sharp)$ satisfies condition (3) of Modules on Sites, Lemma 04H9. To see this take a point p of $X_{étale}$. By Lemma 04K6 p corresponds to a geometric point $\overline{x}$ of X. By Lemma 04K2 the point $q = f_{small} \circ p$ corresponds to the geometric point $\overline{y} = f \circ \overline{x}$ of Y. Hence the assertion we have to prove is that the induced map of étale local rings

$$\mathcal{O}_{Y,\overline{y}} \longrightarrow \mathcal{O}_{X,\overline{x}}$$

is a local ring map. You can prove this directly, but instead we deduce it from the corresponding result for schemes. To do this choose a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow[\psi]{} & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U and V are schemes, and the vertical arrows are surjective étale (see Spaces, Lemma 02X1). Choose a lift $\overline{u} : \overline{x} \to U$ (possible by Lemma 04JX). Set $\overline{v} = \psi \circ \overline{u}$. We obtain a commutative diagram of étale local rings

$$\begin{array}{ccc} \mathcal{O}_{U,\overline{u}} & \longleftarrow & \mathcal{O}_{V,\overline{v}} \\ \uparrow & & \uparrow \\ \mathcal{O}_{X,\overline{x}} & \longleftarrow & \mathcal{O}_{Y,\overline{y}}. \end{array}$$

By Étale Cohomology, Lemma 04I5 the top horizontal arrow is a local ring map. Finally by Lemma 04KF the vertical arrows are isomorphisms. Hence we win. □